Protocols - Express Immobilised Metal Affinity Chromatography (DHAK)

TEMPLATE · IGEM 2026 CONTRIBUTION

What you're going to get / What do you expect

*This protocol is a small-scale, batch version of the IMAC purification of His-tagged **DHAK** from clarified cell-free extract (CFE) on Co^{2+}-IDA-agarose. It runs in a single 1.5 mL Eppendorf tube instead of a packed column. The protocol is **analytical, not preparative**. It is a quick check, using only 25–50 µL of CFE, that DHAK binds the Co^{2+}-IDA resin and can be recovered from it, before you commit a larger sample to the full column protocol (Protocol_ Immobilised Metal Affinity Chromatography). The bound protein is released by denaturation with SDS loading buffer and heat, so the final sample cannot be used for activity assays. It is used only for SDS-PAGE analysis (Laemmli, 1970). If the method works, a band at the expected molecular weight of DHAK should appear in the elution lane and be clearly enriched compared with the starting CFE.*

Materials list

Reagents

- Clarified CFE containing His-tagged DHAK.
- Co^{2+}-IDA-agarose resin, supplied as a suspension.
- Sodium phosphate, dibasic: $Na_2HPO_4 \cdot 2H_2O$ (177.99 g/mol).
- 4× SDS-PAGE loading buffer.
- Ultrapure water (Milli-Q).

Consumables

- 1.5 mL Eppendorf tubes.

Equipment

- Benchtop microcentrifuge.
- Thermoblock (heating block).
- Orbital palte shaker with a foam adaptor.

Steps

A. Buffer preparation:

A single buffer is used for both equilibration and washing: **50 mM sodium phosphate, pH 8.0**, prepared in Milli-Q water, with no NaCl or imidazole. This protocol only aims to confirm that the protein binds the resin, so purity is not a priority, and no additional components are needed.

No Elution Buffer is needed either, because the protein is released from the resin by SDS and heat (Section E).

Note: Adjust the pH after all components have dissolved and before making up to the final volume.

B. Resin preparation and equilibration:

1. Homogenise the Co^{2+}-IDA-agarose suspension by gentle inversion until no sediment remains.
2. Transfer ~0.5 mL of suspension (or slightly less) into a 1.5 mL Eppendorf tube. This volume leaves enough free space in the tube to wash and equilibrate the resin.
3. Pellet the resin with a few short pulses in a benchtop microcentrifuge, then carefully remove and discard the supernatant (the storage solution) with a micropipette, without disturbing the pellet.

4. Add 1 mL of Milli-Q water, resuspend the resin by gentle pipetting, pellet it again with short pulses and discard the supernatant.
5. Add 1 mL of buffer, resuspend, pellet again with short pulses and discard the supernatant. The resin is now equilibrated.

When removing supernatants, always leave a small volume of liquid above the pellet so that you do not aspirate or dry the resin. ⚠️

C. Sample binding:

1. Set aside an aliquot of each CFE ***(Load, L)*** for SDS-PAGE.
2. Transfer ~25 µL of the equilibrated resin into a new Eppendorf tube for each CFE to be tested. Gently resuspend the resin in its residual buffer first.
3. Add 25–50 µL of CFE to each tube.
4. Close the tubes and mix gently by pipetting up and down, to maximise contact between the resin and the protein.
5. Incubate for ~10 min with gentle agitation on an orbital plate shaker, so that the His-tagged protein can bind the Co^{2+} sites.

D. Washing:

1. Pellet the resin with short pulses in the microcentrifuge.
2. Transfer the supernatant to a new labelled Eppendorf tube and keep it as the **Flow-through (FT),** if you want to analyse it on the gel. It contains the CFE proteins that did not bind the resin.
3. Add ~100 µL of buffer, resuspend the resin gently, pellet it again and discard the supernatant (***Wash, W***; keep it if you want to analyse it on the gel).

E. Elution with SDS and heat:

1. Add 20 µL of Milli-Q water and 4× SDS-PAGE loading buffer to the washed resin, to a 1× final concentration.
2. Incubate in the thermoblock at 95 °C for 10 min, as for the CFE samples (Laemmli, 1970). The SDS and the heat denature the protein, so it unfolds and is released from the Co^{2+} sites on the resin. Because the protein is denatured, this sample cannot be used for activity assays.
3. Just before loading the gel, pellet the resin with a short spin so that it collects at the bottom of the tube. Load only the supernatant (***Elution, E***), so that no resin enters the gel well.
4. Prepare the Load fraction (and the **FT and W** fractions, if kept) with loading buffer following Protocol_SDS_PAGE. Run all fractions side by side on the same gel to see where the protein ends up.

Troubleshooting

- **Resin lost during supernatant removal:** Let the resin settle fully, pipette slowly from the top of the liquid, and leave a small volume above the pellet.

- **Resin in the gel well (distorted or smeared lanes):** Always spin the sample again just before loading and take the supernatant from the top.